Japanese contrast sluicing and pair-list questions

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In this squib we investigate contrast sluicing in Japanese with multiple *wh* questions antecedents. Contrast sluicing in such cases presents a puzzling asymmetry, which reflects the differing status of the two *wh*-phrases in pair-list multiple *wh* questions (Kuno, 1982, Dayal, 1996; a.o.). We will argue that this asymmetry can be explained through careful consideration of the syntax/semantics of pair-list questions, together with the common movement-and-deletion approach to the derivation of contrast sluicing (see e.g. Merchant, 2001) and the possibility of argument ellipsis in Japanese.

1 Contrast Sluicing with Multiple *Wh* Questions

Example (1) illustrates a basic case of contrast sluicing in Japanese. We refer to the *wh*-phrase *dono zasshi* 'which magazine' in B as a *remnant*, which contrasts in its nominal domain with a corresponding *wh*-phrase *dono hon* 'which book' in the antecedent clause A. B is interpreted as another embedded question, 'which magazine Taro read.'

(1) **Contrast sluicing in Japanese:**

Watashi-wa [$_A$ Taro-ga [dono HON]-o yonda ka]-wa shitteiru.
I-TOP Taro-NOM which book-ACC read Q -TOP know

Shikashi, *pro* [$_B$ [dono ZASSHI]-o ka]-wa shir-anai.
however which magazine-ACC Q -TOP know-NEG

'I know [which BOOK Taro read]. But I don't know [which MAGAZINE (Taro read)].'

The empirical focus of this paper will be contrast sluicing involving an antecedent *wh*-question with two *wh*-phrases. Contrast sluicing with a single remnant *wh*-phrase is then

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grammatical when the remnant corresponds to and contrasts with the antecedent's lower *wh*, but is ungrammatical when it corresponds and contrasts with the antecedent's higher *wh*. We will be specifically interested in the availability of so-called *pair-list* interpretations for both the antecedent question (A) and for the sluice clause (B). We will discuss pair-list interpretation and its syntax/semantics in more detail in section 2.

We begin with example (2) where contrast sluicing targets the lower *wh*-phrase. A single remnant *wh*-phrase in B, *dono higeki-o* 'which tragedy' contrasts with the lower *wh*-phrase in the antecedent clause A, *dono kigeki-o* 'which comedy.' The example is grammatical with both A and B interpreted as pair-list questions: that is, they are questions that are answered by a list of (student, comedy) or (student, tragedy) pairs that addresses each student in the context. The context in (2) ensures the availability of the intended pair-list readings for both A and B. The higher *wh*-phrase from A, *dono seito-ga* 'which student,' can also be repeated in B with no difference in acceptability.

(2) **[$_{\mathrm{A}}$ ... wh_1 ... wh_2 ...] ... [$_{\mathrm{B}}$ (wh_1) wh'_2]:**

Context: In the dramatic literature class, each student gives a class presentation on one comedy and one tragedy. After the first class, the students met and chose which ones they'd like to present, but the forgetful TA failed to remember the arrangement.

TA-wa	[$_{\mathrm{A}}$ [dono	seito]-ga	[dono	KIGEKI]-o	eranda	ka]-wa	oboeteiru.
TA-TOP	which	student-NOM	which	comedy-ACC	chose	Q-TOP	remember

Shikashi,	*pro* [$_{\mathrm{B}}$ ([dono	seito]-ga)	[dono	HIGEKI]-o	ka]-wa	wasureta.	
however	which	student-NOM	which	tragedy-ACC	Q-TOP	forgot	

'The TA remembers [$_{\mathrm{A}}$ which student chose which COMEDY]. But they forgot [$_{\mathrm{B}}$ which student chose which TRAGEDY].'

Next we consider example (3) below, where the higher *wh*-phrases contrast. The example is grammatical with the intended pair-list interpretation if both *wh*-phrases are pronounced in the sluice clause B, but it is ungrammatical if the noncontrasting lower *wh*-phrase is not repeated.[1] The context here again supports the intended pair-list interpretations for both A and B, which must address each undergrad and each grad student, respectively. This differs from the behavior in example (2) above, where contrast sluicing with only the one, contrasting *wh*-phrase in the remnant was grammatical with the intended pair-list reading.

(3) **[$_A$ ... wh_1 ... wh_2 ...] ... [$_B$ wh'_1 *(wh_2)]:**

Context: The students also discussed when they will give their class presentations. The forgetful TA failed to completely remember this schedule as well.

TA-wa [$_A$ [dono GAKUBUSEI]-ga [dono hi]-o eranda ka]-wa oboeteiru.
TA-TOP which undergrad-NOM which day-ACC chose Q-TOP remember

Shikashi, *pro* [$_B$ [dono INSEI]-ga *([dono hi]-o) ka]-wa wasureta.
however which grad student-NOM which day-ACC Q-TOP forgot

'The TA remembers [$_A$ which UNDERGRAD chose which day]. But they forgot [$_B$ which GRAD STUDENT chose which day].'

The pattern of grammaticality from the examples above is summarized in (4) below. Note that we concentrate here on examples without scrambling in the A clauses. We will discuss the effects of scrambled *wh*-phrases below.

(4) **Summary:**

a. ^{ok} [$_A$ ... wh_1 ... wh_2 ...] ... [$_B$ wh'_2] (2)

b. * [$_A$ ... wh_1 ... wh_2 ...] ... [$_B$ wh'_1] (3)

c. ^{ok} [$_A$ ... wh_1 ... wh_2 ...] ... [$_B$ wh_1 wh'_2] (2)

d. ^{ok} [$_A$ ... wh_1 ... wh_2 ...] ... [$_B$ wh'_1 wh_2] (3)

The summary in (4) immediately leads to two observations. First, the contrast between (4a) and (4b) shows that the two *wh*-phrases in a pair-list multiple *wh* question differ in their status in some way. Second, assuming that deletion is involved in the derivation of contrast sluicing, it is not simply the case that all noncontrastive material is deleted. If that were the case, a structure with a contrasting, higher *wh*-phrase wh'_1 should result in the surface structure in (4b), with the (4d) option dispreferred due to repetition of wh_2. Instead, the observed pattern shows that the

[1] Example (3) without the lower *wh*-phrase in B is however grammatical under another reading, where the antecedent clause B 'which undergrad chose which day' is interpreted as a single-pair question, with the remnant in B interpreted as the single *wh*-question 'which grad student chose *that day* (that an undergrad also chose).' Again, here we concentrate on the availability of pair-list interpretations.

remnant corresponding to the lower *wh*-phrase must always be pronounced in contrast sluicing between pair-list questions. Our proposal will address each of these observations.

2 Two Paths to Pair-List Interpretation

The puzzling asymmetry observed in contrast sluicing with multiple *wh* antecedents, which we first reported in Mizuno and Erlewine 2018 and is summarized in (4), can be explained through consideration of the syntax/semantics of pair-list questions, combined with existing accounts for the derivation of (contrast) sluicing.

We first elaborate further on the nature of pair-list questions. Pair-list multiple *wh* questions request an answer in the form of a list of (x,y) pairs, but they also impose further requirements on their answers. A complete answer to a pair-list *wh*-question with domains X and Y must exhaustively address each value in one of the domains, X, mapping each value in X to a value in Y (Dayal 1996; see also note 4 below). The domain X which must be exhaustively addressed is called the "sorting key" following Kuno 1982.[2]

In the pair-list interpretations of Japanese multiple *wh* questions, the higher *wh*-phrase serves as the sorting key (Kuno and Takami, 1993:116ff; Kitagawa, Roehrs, and Tomioka, 2004). For instance, the pair-list reading of (5) is "sorted" on the higher *wh*-phrase with domain 'student,' and expects an answer which mentions a unique book for each student.

(5) **Pair-list multiple *wh* question sorted on 'student':**

[Dono seito]-ga	[dono hon]-o	yonda	no?
which student-NOM	which book-ACC	read	Q

'Which student read which book?'

Scrambling one *wh*-phrase over another *wh*-phrase generally results in the surface-higher *wh*-phrase then becoming the sorting key.[3] For instance, consider example (6) below which is based on (5) but with the object 'which book' scrambling over the subject 'which student.' The most accessible pair-list reading of (6) takes the sorting key to be books, thereby expecting that each book was read by some student. We will specifically discuss the effects of scrambling in section 5 below, and concentrate on examples without scrambled *wh*-phrases until then.

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[2] Relatedly, Kishimoto (2024) demonstrates that the speaker must have an idea of the possible values in the sorting key domain X, but not of the possible values in Y.

[3] Hagstrom (1998:74–76) reports that rearranging the order of *wh*-phrases by scrambling makes pair-list readings

(6) **Pair-list multiple *wh* question sorted on 'book,' via scrambling from (5):**

[Dono hon]-o$_i$ [dono seito]-ga t_i yonda no?
which book-ACC which student-NOM read Q
'Which book did which student read?'

Pair-list questions can also be expressed using a single *wh*-phrase and a universal quantifier, with the quantifier then serving as the sorting key (see e.g. Chierchia, 1993, Dayal, 2002). See the examples in (7) below, whose embedded questions express pair-list questions that parallel the multiple *wh* questions in (5) and (6) above.[4]

(7) **Quantifier–*wh* questions with pair-list interpretation:**

a. [[Sorezore-no seito]-ga [dono hon]-o yonda ka] oshiete kudasai.
each student-NOM which book-ACC read Q tell please
'Tell me [which book each student read].'

b. [[Sorezore-no hon]-o$_i$ [dono seito]-ga t_i yonda ka] oshiete kudasai.
each book-ACC which student-NOM read Q tell please
'Tell me [which student read each book].'

Many works on the compositional syntax/semantics of questions with pair-list interpretation have converged on the generalization that the sorting key must take scope over the (other) *wh*-phrase at LF. For instance, for quantifier–*wh* pair-list questions, Szabolcsi (1997) and Krifka (2001) propose that the universal quantifier serves as the sorting key by scoping out of the *wh*-question. Kitagawa, Roehrs, and Tomioka (2004) adapt this approach for pair-list multiple *wh* questions as well, with the sorting key *wh*-phrase taking scope over the other *wh*-phrase's question speech act. Alternatively, a pair-list question may denote a set of simplex question denotations following Hagstrom (1998:ch. 6) and Büring (2003), which is sometimes called a "family of questions" denotation. Fox (2012) and Kotek (2014, 2019) derive such family of

less accessible, but see Kitagawa, Roehrs, and Tomioka 2004:229–231 for arguments against this description.

[4] We embed these questions under 'tell [me]' here as the pair-list interpretations are more accessible for some speakers when embedded. Other types of universal quantifiers can also yield the same effect, subject to some variation. For instance, Hoji (1985:270) reports that a *wh-mo* universal subject does not support the pair-list interpretation, but Choi (2002:169) reports some attested variation on this point.

An anonymous reviewer reports an interpretational difference between corresponding *wh*–*wh* and quantifier-*wh* questions, in that the former but not the latter expects an answer where each value of X corresponds to a *different* value of Y. We ourselves do not detect such a difference between the two types of pair-list questions, and perceive the embedded questions in (7a,b) to be equivalent to (5) and (6), respectively. We will address the contrast reported by this reviewer again in note 8 below.

questions denotations in two different traditions for the composition of *wh*-questions — following Karttunen 1977 and Hamblin 1973, respectively — in both cases requiring the sorting key *wh* to take wide scope at LF. Finally, Comorovski (1996) and Dayal (1996) treat pair-list questions on a par with quantifier–*wh* questions with so-called functional answers, again requiring the sorting key to take scope over the (lower) *wh*-phrase at LF.

To summarize, the two *wh*-phrases in a question with pair-list interpretation do not have the same status: the higher *wh*-phrase is the "sorting key," which is reflected in the structure of a complete answer and the corresponding presuppositions of the question. Pair-list questions can also be expressed with a single *wh*-phrase and a universal quantifier serving as the sorting key. In either case, the sorting key phrase takes wide scope at LF. With this background in place, we now return to the main puzzle of this paper: an asymmetry in patterns of contrast sluicing, summarized in (4) above.

3 Explaining the Asymmetry in Pair-List Contrast Sluicing

Contrast sluicing involves two parallel questions which vary only in a nominal domain. Let the antecedent pair-list question A be a multiple *wh* question with wh_1 ranging over the sorting key domain X and wh_2 ranging over the domain Y. We first consider the case where the B question differs from A only in referring to domain Y′ in place of Y. Following the discussion in the previous section, a pair-list question could be expressed as a multiple *wh* question or as a single *wh* question with a higher quantifier. These two options for B are illustrated schematically in (8) below. $\forall_1$ stands in for a universal quantifier which ranges over the domain of wh_1, X.

(8) $[_{\text{A}}$ ... wh_1 ... wh_2 ...] ... $[_{\text{B}}$ ... $\{wh_1 / \forall_1\}$ ... wh_2' ...]

Now consider sluicing based on the B clauses in (8). For concreteness, we present this analysis in further detail using the movement-and-deletion derivation for contrast sluicing in Japanese (Takahashi, 1994, Fukaya and Hoji, 1999, Fukaya, 2003, 2007, 2012, Abe, 2015).[5] Under the movement-and-deletion approach, we assume the possibility of scope-preserving movement to the CP edge of the *wh*-phrase(s) and, in a quantifier–*wh* question, the relevant universal quantifier. See the LF representations in (9). These movements must be overt in the B clause, as they will be followed by deletion of the TP.

[5] See especially Fukaya's work for evidence that contrast sluicing in Japanese is island-sensitive, unlike noncontrastive

(9) <u>LF:</u> $[_{CP}\ \{wh_1 / \forall_1\}\ [\ wh_2^{(\prime)}\ [_{TP} \ldots t_1 \ldots [\ \ldots t_2 \ldots\ ]]]]$

Following these movements in both A and B, their TP nodes will be syntactically and semantically identical, licensing ellipsis (Sag, 1976; a.o.). The TP in B is elided under identity with the TP in A. This yields the surface structure in (10):

(10) $\ldots [_{CP(B)}\ wh_1/\forall_1\ [\ wh'_2\ \cancel{[_{TP} \ldots t_1 \ldots [\ \ldots t_2 \ldots\ ]]}]]$

This predicts two possible surface patterns of sluicing, with the same interpretation: one with multiple *wh*-phrases, attested above in (2) (pattern 4c), and one with a universal quantifier and a *wh*-phrase. This latter possibility is indeed attested, as in (11):

(11) **[$_{A}$ ... *wh*$_1$ ... *wh*$_2$ ...] ... [$_{B}$ $\forall_1$ *wh*$'_2$]:** (based on (2))

TA-wa [$_A$ [dono seito]-ga [dono KIGEKI]-o eranda ka]-wa oboeteiru.
TA-TOP which student-NOM which comedy-ACC chose Q-TOP remember

Shikashi, *pro* [$_B$ [sorezore-no seito]-ga [dono HIGEKI]-o ka]-wa wasureta.
however each student-NOM which tragedy-ACC Q-TOP forgot

'The TA remembers [$_A$ which student chose which COMEDY]. But they forgot [$_B$ which TRAGEDY each student chose].'

We furthermore propose that the universal quantifier in (11) can undergo the independent operation of *argument ellipsis* (AE). AE refers to the availability of null arguments in Japanese which receive quantificational interpretations unpredicted by referential null *pro* (Oku, 1998, Saito, 2007, Takahashi, 2008, Mizuno, 2025; a.o.). AE of the universal quantifier in (11), schematized in (12) below,[6] results in the attested pattern of contrast sluicing observed above in (2), where the remnant B includes only one *wh*-phrase, corresponding to and contrasting with the lower *wh*-phrase of the antecedent pair-list question A (pattern 4a).[7] We propose that single

sluicing, and Abe 2015:21–26, 93–98 for subsequent discussion. See also Merchant 2008 and Griffiths and Lipták 2014 on the island-sensitivity of contrast sluicing cross-linguistically.

6 As noted by a reviewer, what we describe as a missing universal quantifier $\forall_1$ in (12) could potentially be a plural null *pro* or plural E-type pronoun, which is then interpreted distributively with wide-scope. See especially Tomioka 2014 on the challenges of distinguishing many cases of apparent AE from the use of E-type pronouns. See also Johnston 2023 for recent discussion of pair-list questions with plural definite sorting keys.

7 The structure here in (12) is in line with recent arguments in Mizuno 2025, made on independent grounds, which show that targets of AE first undergo movement (topicalization) to a CP edge (also following Fujiwara, 2022:ch. 2) and that, in *wh*-questions, AE can only target phrases that are higher than the *wh*-phrase.

remnant contrast sluicing with a pair-list antecedent always reflects an underlying quantifier–*wh* question, rather than a *wh*–*wh* question.[8]

(12) ... $[_{\text{CP(B)}}$ $\forall_T$ [wh'_2 $\cancel{[_{\text{TP}} \ldots t_1 \ldots [\ldots t_2 \ldots]]}$]]

Next we turn to pair-list questions with contrasting sorting key domains, X vs X′. Again, with a multiple *wh* question antecedent A, the pair-list question B could involve two *wh*-phrases or a single *wh*-phrase with a higher quantifier:

(13) $[_{\text{A}} \ldots wh_1 \ldots wh_2 \ldots] \ldots [_{\text{B}} \ldots \{wh'_1 \,/\, \forall'_1\} \ldots wh_2 \ldots]$

Now consider the options for contrast sluicing based on each of these B clauses in (13). Following movement of the remnant phrases, deletion of TP results in (14):

(14) ... $[_{\text{CP(B)}}$ $\{wh'_1 \,/\, \forall'_1\}$ [wh_2 $\cancel{[_{\text{TP}} \ldots t_1 \ldots [\ldots t_2 \ldots]]}$]]

(14) predicts two possible surface patterns of sluicing: one with both *wh*-phrases, attested above in (3) (pattern 4d), and one with a universal quantifier and a *wh*-phrase. This latter possibility too is attested:

(15) **$[_{\text{A}} \ldots wh_1 \ldots wh_2 \ldots] \ldots [_{\text{B}}\ \forall'_1\ wh_2]$:** (based on (3))

TA-wa $[_{\text{A}}$ [dono GAKUBUSEI]-ga [dono hi]-o eranda ka]-wa oboeteiru.
TA-TOP which undergrad-NOM which day-ACC chose Q-TOP remember

Shikashi, *pro* $[_{\text{B}}$ [sorezore-no INSEI]-ga [dono hi]-o ka]-wa wasureta.
however each grad student-NOM which day-ACC Q-TOP forgot

'The TA remembers $[_{\text{A}}$ which UNDERGRAD chose which day]. But they forgot $[_{\text{B}}$ which day each GRAD STUDENT chose].'

In contrast to the case with contrasting Y domains as in (11) above, though, the remnant in (15) cannot be further reduced through AE. AE can target universal quantifiers — as illustrated

8 Recall from note 4 above that an anonymous reviewer reports a subtle difference in interpretation between corresponding *wh*–*wh* and quantifier–*wh* pair-list questions. This reviewer also reports the same difference between the two variants of example (2) above, such that the sluice clause with both *wh*-phrases pronounced (pattern 4c) yields a *wh*–*wh* question interpretation, whereas the variant with only the lower *wh*-phrase remnant (pattern 4a) leads to a quantifier–*wh* question interpretation. (Concretely, they report that the former but not the latter comes with an expectation that the students chose different tragedies. See note 4.) As the reviewer notes, this pattern of judgments (for those speakers that detect this difference between the two types of pair-list questions, such as this reviewer but not ourselves) further strengthens the motivation for our proposal here, where the single *wh*-phrase contrast sluice actually reflects an underlying quantifier–*wh* structure as in (12), rather than an underlying *wh*–*wh* structure.

schematically in (12) above to yield pattern (4a) — but in this case (15), the universal quantifier $\forall'_1$ includes contrastive material and cannot be deleted (Fiengo and Lasnik, 1972). It is also impossible to elide the noncontrastive *wh*-phrase wh_2 'which day' in (15), due to what Saito (2017:723) describes as "the well-known fact that interrogative *wh*-phrases resist argument ellipsis," first discussed by Sugisaki (2012) and Ikawa (2013):[9]

(16) **Argument ellipsis cannot target *wh*-phrases:**

a. Hanako-wa [dare-ga KYOTO-e itta ka] shitteiru.
Hanako-TOP who-NOM Kyoto-to went Q know
'Hanako knows [who went to KYOTO].'

b. * Taro-wa [~~dare-ga~~ TOKYO-e itta ka] shitteiru.
Taro-TOP who-NOM Tokyo-to went Q know
Intended: 'Taro knows [who went to TOKYO].'

We have now derived the puzzling asymmetry presented in section 1. The summary in (4) is repeated together with illustrations for the structures underlying the remnants in B.

[9] An important exception is the fact that (argument) ellipsis may target an entire *wh*-question, including its *wh*-phrase(s) as well as its complementizer, as summarized in Sakamoto 2019:128–129. For instance, (16a) could be followed by *Taro-wa* Δ *shira-nai* 'Taro doesn't know Δ', eliding the entire embedded question. See also Li 2016 and Hiraiwa and Kobayashi 2020 for related evidence and discussion.

Saito (2017) describes this effect as due to a restriction that "argument ellipsis does not apply to a phrase that forms an operator-variable chain" (p. 724). We note as well that, assuming that the interpretation of in-situ *wh*-phrases involves the projection of Rooth-Hamblin alternatives (as per Hagstrom, 1998, Shimoyama, 2001, 2006, Beck, 2006, Kotek, 2014, 2019, Erlewine and Kotek, 2018, Uegaki, 2018, Oda, 2021, Erlewine, 2025, among others), this ungrammaticality of eliding a *wh*-phrase without eliding its entire question can be unified with a generalization regarding focus association, that a focus (or focus-containing phrase) cannot be elided while stranding the corresponding focus-sensitive operator (Beaver and Clark, 2008:177):

(i) From Bassi, Hirsch, and Trinh 2022:817:
A: I only know he brought WHITE_F wine. What about you?
B: * I only know he did ~~[$_{\text{VP}}$ bring WHITE_F wine]~~, too.

These proposals also explain the ungrammaticality of eliding a VP containing an in-situ *wh* in an English multiple *wh* question, which Abels and Dayal (2023) leave as an open puzzle in their footnote 13.

(17) **Deriving the pattern of contrast sluicing in (4):**

a. ^{ok} $[_{A}$ ... wh_1 ... wh_2 ...] ... $[_{B}$ wh'_2]

B = $\forall$–*wh* pair-list question with argument ellipsis of $\forall_1$

= $[_{CP}$ $\cancel{\forall_1}$ [wh'_2 $\cancel{[_{TP} \ldots t_1 \ldots [\ldots t_2 \ldots]]}$]]

b. * $[_{A}$ ... wh_1 ... wh_2 ...] ... $[_{B}$ wh'_1]

B = multiple *wh* pair-list question with argument ellipsis of wh_2

= $[_{CP}$ wh'_1 [$\cancel{wh_2}$ $\cancel{[_{TP} \ldots t_1 \ldots [\ldots t_2 \ldots]]}$]]

ungrammatical as *wh*-phrases cannot be elided (Sugisaki, 2012, Ikawa, 2013; a.o.)

c. ^{ok} $[_{A}$ ... wh_1 ... wh_2 ...] ... $[_{B}$ wh_1 wh'_2]

B = multiple *wh* pair-list question, no argument ellipsis

= $[_{CP}$ wh_1 [wh'_2 $\cancel{[_{TP} \ldots t_1 \ldots [\ldots t_2 \ldots]]}$]]

d. ^{ok} $[_{A}$ ... wh_1 ... wh_2 ...] ... $[_{B}$ wh'_1 wh_2]

B = multiple *wh* pair-list question, no argument ellipsis

= $[_{CP}$ wh'_1 [wh_2 $\cancel{[_{TP} \ldots t_1 \ldots [\ldots t_2 \ldots]]}$]]

This full pattern is derived by maintaining the generalization that *wh*-phrases are never deleted under sluicing, here implemented through obligatory scope-preserving movement of *wh*-phrases and relevant quantifiers to the CP edge, followed by deletion of TP. Key to deriving the observed asymmetry is the recognition that any particular pair-list question can be constructed using two *wh*-phrases or a single *wh*-phrase with a distributive quantifier. Quantifiers, unlike *wh*-phrases, can undergo argument ellipsis. The apparent difference between higher and lower *wh*-phrases in the pattern of contrast sluicing in (4/17) is due to the possibility of employing a quantifier in place of a higher *wh*-phrase, but the inability of such replacement for lower *wh*-phrases.

4 Other Patterns of Contrast Sluicing with Pair-List Questions

As a reviewer notes, our discussion correctly predicts the possibility of various other forms of contrast sluicing with a quantifier–*wh* pair-list question antecedent and remnants of *wh*–*wh* or quantifier–*wh* form. When there is contrast between the lower arguments (here, ‘comedy’ versus ‘tragedy’ in (18)), the higher, sorting key argument may be included optionally in the sluice, as a universal quantifier or as a *wh*-phrase:

(18) **$[_{\mathrm{A}}$ ... $\forall_1$ ... wh_2 ...] ... $[_{\mathrm{B}}$ $(wh_1\ /\ \forall_1)$ wh'_2]:** (based on (2))

TA-wa [$_{\mathrm{A}}$ [sorezore-no seito]-ga [dono KIGEKI]-o eranda ka]-wa oboeteiru.
TA-TOP each student-NOM which comedy-ACC chose Q-TOP remember

Shikashi, *pro* [$_{\mathrm{B}}$ ([dono/sorezore-no seito]-ga) [dono HIGEKI]-o ka]-wa wasureta.
however which/each st.-NOM which tragedy-ACC Q-TOP forgot

'The TA remembers [$_{\mathrm{A}}$ which COMEDY each student chose]. But they forgot [$_{\mathrm{B}}$ which student chose which TRAGEDY / which TRAGEDY each student chose].'

When the higher, sorting key arguments contrast ('undergrad' versus 'grad student' in (19)), the (lower) *wh*-phrase must be included in B, even though it is not contrastive:

(19) **$[_{\mathrm{A}}$ ... $\forall_1$ ... wh_2 ...] ... $[_{\mathrm{B}}$ $\{wh'_1\ /\ \forall'_1\}$ *(wh_2)]:** (based on (3))

TA-wa [$_{\mathrm{A}}$ [sorezore-no GAKUBUSEI]-ga [dono hi]-o eranda ka]-wa oboeteiru.
TA-TOP each undergrad-NOM which day-ACC chose Q-TOP remember

Shikashi, *pro* [$_{\mathrm{B}}$ [dono/sorezore-no INSEI]-ga *([dono hi]-o) ka]-wa wasureta.
however which/each g.s.-NOM which day-ACC Q-TOP forgot

'The TA remembers [$_{\mathrm{A}}$ which day each UNDERGRAD chose]. But they forgot [$_{\mathrm{B}}$ which GRAD STUDENT chose which day / which day each GRAD STUDENT chose].'

The pattern in (18–19) accords with our core generalization as in (4/17) and is straightforwardly accounted for under our analysis. Each logical pair-list question can take two syntactic forms — a multiple *wh* question and a quantifier–*wh* question — and, in contrast sluicing between pair-list questions, each pair-list question can take either syntactic form. In all of these combinations, the grammatical patterns of contrast sluicing are accurately predicted by the generalization that *wh*-phrases cannot be deleted under sluicing — implemented here by the movement-and-deletion approach — together with the possibility of argument ellipsis applying to quantifiers but not *wh*-phrases in Japanese.

5 The Effects of Scrambling

Recall from our discussion of examples (5–6) above that scrambling one *wh*-phrase over another can change which serves as the sorting key. Dayal (1996:113–114) and Kishimoto (2024:23) describe this effect of scrambling as obligatory in Japanese and, as a result, the *surface-higher* *wh*-phrase always serves as the sorting key. As predicted, then, contrast sluicing with a single *wh*-phrase remnant that contrasts with the antecedent's underlyingly higher but *surface-lower* *wh*-phrase is grammatical, as in (20). The context here ensures that the presupposition of the pair-list question with 'day' as the sorting key is met.

(20) **[$_A$ ... wh_1 ... wh_2 ... t_1 ...] ... [$_B$ (wh_1) wh'_2]:**

Context: In each class, one undergraduate student and one graduate student will give presentations. The forgetful TA failed to completely remember this schedule.

TA-wa [$_A$ [dono hi]-o$_i$ [dono GAKUBUSEI]-ga t_i eranda ka]-wa oboeteiru.
TA-TOP which day-ACC which undergrad-NOM chose Q-TOP remember

Shikashi, *pro* [$_B$ ([dono hi]-o) [dono INSEI]-ga ka]-wa wasureta.
however which day-ACC which grad student-NOM Q-TOP forgot

'The TA remembers [$_A$ which day which UNDERGRAD chose]. But they forgot [$_B$ which day which GRAD STUDENT chose].'

Next, we consider contrast sluicing where a single remnant contrasts with the *surface-higher* *wh*-phrase. On the surface, this would be an instance of pattern (4b/17b) above, which is ungrammatical; see (3). However, as noted by a reviewer, this configuration appears to be grammatical when the antecedent question involves *wh*-scrambling:

(21) **Apparent counterexample to (4b/17b) with a scrambled antecedent:**

TA-wa [$_A$ [dono KIGEKI]-o$_i$ [dono seito]-ga t_i eranda ka]-wa oboeteiru.
TA-TOP which comedy-ACC which student-NOM chose Q-TOP remember

Shikashi, *pro* [$_B$ [dono HIGEKI]-o ka]-wa wasureta.
however which tragedy-ACC Q-TOP forgot

'The TA remembers [$_A$ which student chose which COMEDY]. But they forgot [$_B$ which student chose which TRAGEDY].' =(2)

However, when we pay closer attention to meaning of these pair-list questions in (21), we see that the counterexample is only apparent. Both of the questions in A and B are sorted on 'students,' equivalent to the interpretation of example (2) above and compatible with the context there which supports a 'student' sorting key interpretation. In other words, the sluicing in (21) is actually of the grammatical type in (4a/17a), with the question in A interpreted as if scrambling had not taken place.

As noted above, the consensus in the literature has been that scrambled *wh*-phrases change the interpretation of pair-list questions in Japanese, so that the surface-higher *wh*-phrase serves as the sorting key. At the same time, it is well known that scrambled quantifiers can introduce scope-ambiguities, reflecting the possibility of reconstruction at LF (see e.g. Hoji, 1985, Saito, 1985). The grammaticality of example (21) where sluicing forces a 'student' sorting key interpretation shows that scrambled *wh*-phrases *can* reconstruct, although such interpretations may otherwise be difficult to access.

6 Conclusion

We began with a puzzling asymmetry in the possible patterns of contrast sluicing between pair-list questions in Japanese with multiple *wh* question antecedents. We proposed that pair-list questions of both the *wh*–*wh* and quantifier–*wh* types may underlie the remnants of sluicing, with the possibility of argument ellipsis targeting quantifiers but not *wh*-phrases deriving the observed asymmetry. The data here highlights the distinct status of the two *wh*-phrases in multiple *wh* pair-list questions, and also constitutes an argument against approaches to contrast sluicing that predict that any and all noncontrastive material can or should be elided.

As a final note, we observe that our analysis also extends to explain a parallel contrast in Japanese noncontrastive sluicing with a pair-list question antecedent. Example (22) gives three different possible sluices, with B intended to be interpreted as equivalent to the pair-list *wh*-question in A with sorting key 'who.' Nishigauchi (1998, 2002) notes that the single sluice with the higher 'who' (22a) is ungrammatical whereas the single sluice with the lower 'what' (22b) is better but "not perfectly acceptable" (Nishigauchi, 2002:97) which we indicate with ? here. The multiple sluice in (22c) is perfectly acceptable.

(22) **Parallel contrast in noncontrastive sluicing:**

(2002:97–98; see also Nishigauchi 1998:139–140)

Hanako-ga [$_A$ dare-ga nani-o motte kuru ka] kimeta. Taro-wa...
Hanako-NOM who-NOM what-ACC bring come Q decided Taro-TOP
'Hanako determined [$_A$ who will bring what]. Taro...'

a. * [$_B$ dare-ga ka] wasureta.
who-NOM Q forgot

b. $^?$ [$_B$ nani-o ka] wasureta.
what-ACC Q forgot

c. [$_B$ dare-ga nani-o ka] wasureta.
who-NOM what-ACC Q forgot
'...forgot [$_B$ who will bring what].'

Our analysis here immediately explains the contrast between Nishigauchi's ungrammatical (22a) and more acceptable (22b). As for the slight unnaturalness of (22b), Nishigauchi hypothesizes that this is due to the fact that ellipsis of the entire embedded question, *Taro-wa* Δ *wasureta*, is also possible (see note 9). Note that eliding the entire embedded question is not possible in our contrast sluicing examples. By concentrating on contrast sluicing as we have done here, we have been able to avoid this confound in Nishigauchi's paradigm and more clearly observe the difference in status between the two *wh*-phrases in pair-list questions.

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